$$E(XY) = \Sigma\Sigma\, xy\, p(x,y)$$

$$= 1 \times 0.25 + 0 \times 0.25 + 0 \times 0.25 + 0 \times 0.25$$

$$= 0.25$$

$$E(X) = \Sigma\, x\, p_i$$

$$= 1 \times 0.5 + 0 \times 0.5$$

$$= 0.5$$

$$E(Y) = \Sigma\, y\, p_j$$

$$= 1 \times 0.5 + 0 \times 0.5 = 0.5$$

$$E(X) \times E(Y) = 0.5 \times 0.5 = 0.25$$

$$E(XY) = E(X)\, E(Y)$$

[It is applicable only when X and Y are independent]

Example 9.18

The independent random variables X and Y have the p.d.f given by

$$f(x) = \begin{cases} 4ax, & 0 \le x \le 1 \\ 0, & otherwise \end{cases}$$

$$f(y) = \begin{cases} 4by, & 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

Prove that $E(XY) = E(X)\, E(Y)$

Solution:

X and Y are independent

$$f(x,y) = f(x) \times f(Y)$$

$$f(x,y) = 4ax \times 4by$$

$$f(x,y) = \begin{cases} 16abxy, & 0 \le x,y \le 1 \\ 0, & otherwise \end{cases}$$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\, dxdy$$

$$= \int_0^1\int_0^1 xy \times 16abxydxdy$$